a) Insufficient crypto magic.

b) Browsers may reject the second cookie.

c) An attacker may be able to overwrite the second cookie.

d) An attacker may be able to guess the second cookie value.

e) An attacker can exploit a timing attack to discover the second cookie value.

## 4.5 Building the Natter login UI

Now that you've got session-based login working from the command line, it's time to build a web UI to handle login. In this section, you'll put together a simple login UI, much like the existing Create Space UI that you created earlier, as shown in figure 4.13. When the API returns a 401 response, indicating that the user requires authentication, the Natter UI will redirect to the login UI. The login UI will then submit the username and password to the API login endpoint to get a session cookie, set the anti-CSRF token as a second cookie, and then redirect back to the main Natter UI.